

COLLEGE CODE: 8203

COLLEGE: AVC COLLEGE OF ENGINEERING

DEPARTMENT: INFORMATION TECHNOLOGY

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ROLL NO: 23IT112

DATE:15/09/2025

Completed the project named as

Phase_2 TECHNOLOGY PROJECT

NAME: Online Quiz Application

SUBMITTED BY,

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TECHSTACKSELECTION:

1.Frontend (View Layer – User Interface):

React.js – for building dynamic quiz interfaces.

Tailwind CSS / Material UI – for responsive and modern UI components.

Axios / Fetch API – to interact with backend REST APIs.

2.Backend (Controller Layer – Business Logic):

Node.js – lightweight, event-driven runtime for server-side execution.

Express.js – to handle REST APIs, routing, and middleware.

JWT Authentication (Optional) – for secure user login & session handling.

3.Database (Model Layer – Data Storage):

MongoDB – NoSQL database for storing quiz questions, options, answers, and user history.

Mongoose ORM – schema-based modeling for MongoDB collections.

4.Deployment & Hosting:

Frontend: Vercel / Netlify (for React hosting).

Backend: Heroku / AWS / Render (for Node.js & Express).

Database: MongoDB Atlas (cloud-based MongoDB).

5.Additional Tools:

Postman / Swagger – for API testing and documentation.

Git & GitHub/GitLab – version control.

Jest / Mocha – for unit testing backend logic.

UISTRUCTURE/APISCHEMA DESIGN:

1. Login/ Register

- Fields: Email, Password, Name
- Buttons: Login, Register

2. Dashboard

- Overview of projects/tasks
- Quick action buttons: Add Project, View Tasks

3.Project Page

- Project list with title, description
- Actions: Add, Edit, Delete Project

4. Task Page

- Task list with title, description, status, due date
- Actions: Add, Edit, Delete Task

5. Profile Page

- View/update user info
- Change password

Project Manager

Sign in to your account or create a new one

Email

Password

Total Projects

3

Total Tasks

5

1 completed

In Progress

2

2 pending

Overdue

4

Need attention

Quick Actions

Get started with these common tasks

+ Add Project

View Tasks

Recent Projects

Your latest projects

Website Redesign

Complete overhaul of company website with modern design and improved UX

2 tasks

Mobile App Development

Develop cross-platform mobile application for iOS and Android

2 tasks

Marketing Campaign

Launch new product marketing campaign across multiple channels

1 tasks

Recent Tasks

Your latest tasks

Design homepage mockup

Due: 2024-01-15

completed

Set up development environment

Due: 2024-01-20

in progress

Conduct market research

Due: 2024-01-25

todo

Implement responsive navigation

Due: 2024-01-18

todo

Profile

Manage your account settings and preferences

Profile Information

Update your personal information and contact details

V

Valli

vallichandrasekar5@gmail.com

Full Name

Valli

Email Address

vallichandrasekar5@gmail.com

Edit Profile

API SCHEMA DESIGN:

Authentication APIs

Method	Endpoint	Request Body	Response	Description
POST userId, token}	/auth/register	{name, email, password}		{message, Register new user
POST	/auth/login	{email, password}	{message, token, user}	Login user
POST	/auth/logout	{token}	{message}	Logout user

Project APIs

Method	Endpoint	Request Body	Response	Description
GET	/projects	-	[[projectId, title, tasks]]	Get all projects
POST	/projects	{title, description}	{message, projectId}	Add new project
PUT updatedProject}	/projects/:id	{title, description}	{message,	Update project
DELETE	/projects/:id	-	{message}	Delete project

Task APIs

Method	Endpoint	Request Body	Response	Description
GET tasks for project	/projects/:id/tasks	-	[[taskId, title, status]]	Get
POST	/projects/:id/tasks	{title, description, dueDate}		Add new task
		{message, taskId}		

PUT updatedTask}	/tasks/:id	{title, description, status}	{message,
	Update task		
DELETE	/tasks/:id	- {message}	Delete task

DATA FLOW:

1. User Interaction (Frontend/UI)

- User enters input (login, create project, add task, update profile).
- Input is validated on the frontend (e.g., required fields, email format).

2. API Communication

- Validated data is sent to the backend through RESTful APIs (JSON format).
- Example: { "title": "Project A", "description": "Web App" }

3. Backend Processing

- Backend service verifies data (auth checks, business rules).
- If valid → passes data to Database Access Layer.
- If invalid → sends error response back to UI.

4. Database Storage

- Data is stored in structured tables:
- User Table: userId, name, email, password (hashed), createdAt
- Project Table: projectId, title, description, ownerId, createdAt
- Task Table: taskId, projectId, title, description, status, dueDate

5. Data Retrieval

- Backend queries database using filters (e.g., all tasks for a project).
- Results are converted into JSON and sent to UI.
- Example: [{ "taskId": 1, "title": "Task A", "status": "Pending" }]

6. UI Display

- Data is displayed in UI components (cards, tables, lists).
- Users see updates instantly (real-time sync possible with WebSockets).

2. Data Handling Best Practices

Validation: Input validated on both frontend and backend.

Security: Passwords stored as hashed (bcrypt/argon2), tokens used for authentication.

Error Handling: Clear error messages returned from API.

Data Integrity: Foreign key relationships (e.g., task belongs to project, project belongs to user).

Consistency: APIs always return JSON responses (success/failure).

Scalability: Pagination used for large datasets (e.g., tasks list).

COMPONENTS/MODULES:

Frontend (UI) Modules

1. Authentication Module

- Login Form
- Registration Form
- Logout Function

2. Dashboard Module

- Overview Cards
- Quick Action Buttons

3. Project Module

- Project List
- Add/Edit/Delete Project Forms

4. Task Module

- Task List
- Add/Edit/Delete Task Forms

5. Profile Module

- View / Update Profile
- Change Password

Backend Modules

1. Authentication Service

- Handles login, registration, logout
- Issues JWT tokens

2. Project Service

- CRUD operations for projects
- Validates user access

3. Task Service

- CRUD operations for tasks
- Assigns tasks to projects/users

4. Database Access Module

- Queries for User, Project, Task tables
- Handles data storage/retrieval

Database Modules

1.User Table

Fields: userId, name, email, password, createdAt

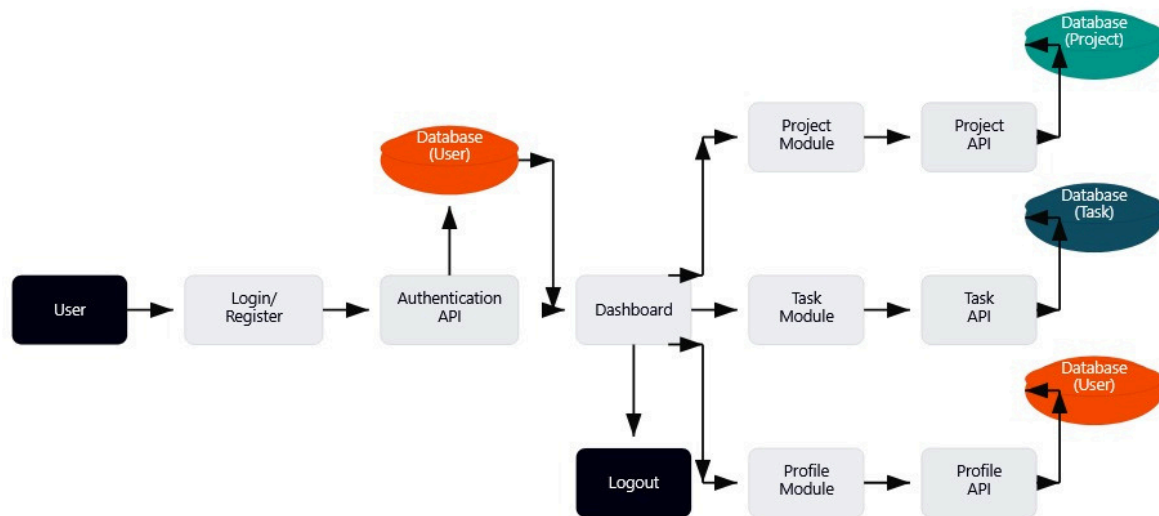
2.Project Table

Fields: projectId, title, description, ownerId

3.Task Table

Fields: taskId, projectId, title, description, status, dueDate

BASIC FLOW DIAGRAM:



BASIC FLOW STEPS:

1. User opens the application

- Chooses to Login or Register

2. Authentication

- Sends login/register data to Authentication API
- Backend verifies credentials
- Returns success/failure response

3. Dashboard

- User sees overview of projects/tasks
- Can click on Project to view or manage

4. Project Management

- Add/Edit/Delete project via Project API
- Backend updates Project Table in database

5. Task Management

- Add/Edit/Delete task via Task API
- Backend updates Task Table in database

6. Profile Management

- User updates profile/password via Profile API
- Backend updates User Table in database

7. Logout

- User logs out, backend invalidates token